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Effect of *Cordyceps militaris* on formation of short-chain fatty acids as postbiotic metabolites

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Abstract

The aim of the current study was to determine the growth-promoting-effect of *Cordyceps militaris*, known as a medicinal mushroom, on *Lactobacillus casei* and *Lactobacillus acidophilus*. To evaluate the best growth-promoting activity of the test substrates including glucose, inulin, and at different concentrations of *C. militaris* (0.5%, 1%, and 2%), the cell counts, optical density (OD), prebiotic activity scores, and postbiotics (lactic, acetic, butyric, and propionic acids) were determined. The highest cell count was found for *L. casei* in media containing 0.5% *C. militaris* and for *L. acidophilus* in media containing 1% *C. militaris*. In the case of both strains, the OD values of the medium with *C. militaris* (1%) and (2%) increased similar to those of glucose. The prebiotic activity scores for both strains were positive. The concentration of lactic acid ranged from 0.56 to 8.07 g L⁻¹ for *L. casei* and 0.82 to 5.38 g L⁻¹ for *L. acidophilus*. Moreover, propionic acid was the highest among short-chain fatty acids (SCFAs) produced by both strains. According to the results of the present study, the tested *Lactobacillus* species can utilize *C. militaris* as carbon source and is able to form postbiotics in the media.

Keywords: Cordyceps militaris; Lactobacillus; SCFAs; postbiotics; prebiotic; probiotic.

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